📊 Statistical Analysis Report

Dataset: Customer Churn (500 records)
Date: April 27, 2026

1. Executive Summary

This report presents a comprehensive statistical analysis of customer churn data. Using descriptive statistics, hypothesis testing, confidence intervals, and logistic regression, the study identifies key drivers of churn and provides actionable business recommendations.

Key Findings:

- Customers with **shorter tenure** and **higher monthly charges** are significantly more likely to churn.
- **Contract type** is the strongest predictor of churn, month-to-month contracts show the highest risk, while one-year and two-year contracts drastically reduce churn.
- **Payment method** does not independently predict churn.
- Logistic regression achieved a **Pseudo R^2 of 0.73**, indicating excellent explanatory power.

Business Implication: Retention strategies should prioritize short-tenure, high-charge customers and incentivize long-term contracts.

2. Descriptive Statistics

The dataset contains 500 customers with attributes including tenure, monthly charges, total charges, contract type, payment method, and churn status.

Metric	**Mean**	**Std Dev**	**Min**	**Max**	**Mode**
Tenure (months)	36.5	20.7	1	71	3
Monthly Charges	113.6	51.8	20	199	115
Total Charges	4237.9	2260.6	159	7992	4023

Interpretation:

- Customers typically stay for ~3 years, but churn is concentrated among those with tenure
- Monthly charges vary widely, with higher charges linked to churn.
- Total charges reflect cumulative spending, strongly correlated with tenure.

3. Data Distribution Analysis

- **Shapiro-Wilk Test (MonthlyCharges):** $p \approx 1.1e-11$ → Reject normality assumption.
- **Implication:** Monthly charges are not normally distributed, justifying the use of non-parametric tests and logistic regression.

4. Hypothesis Testing

1. **T-test (MonthlyCharges vs Churn):**
    - $p = 0.016$ → Significant difference.
    - Customers who churn pay higher monthly charges.
2. **Chi-square (Contract vs Churn):**
    - $p \approx 9.6e-07$ → Highly significant.
    - Contract type strongly influences churn.
3. **Chi-square (PaymentMethod vs Churn):**

- $p = 0.144$ → Not significant.
- Payment method does not independently explain churn.

5. Confidence Intervals

- **95% CI for MonthlyCharges:** ₹109 – ₹118
  Interpretation: The true average monthly charge lies within this range with high confidence. This provides a benchmark for pricing strategies.

6. Logistic Regression Results

Model Fit:

- Pseudo $R^2 = 0.73$ → Strong explanatory power.
- Warning: Quasi-separation indicates some combinations (e.g., long tenure + two-year contract) almost perfectly predict no churn.

Variable	Coefficient	p-value	Interpretation
Tenure	-0.368	<0.001	Longer tenure reduces churn risk significantly
MonthlyCharges	+0.018	0.003	Higher charges increase churn likelihood
Contract (One year)	-3.373	<0.001	Strongly reduces churn
Contract (Two year)	-3.021	<0.001	Strongly reduces churn
PaymentMethod (Credit Card)	+0.852	0.200	Not statistically significant
PaymentMethod (E-Check)	+1.029	0.158	Not statistically significant

7. Business Insights

- **Tenure:** Short-tenure customers are at highest risk. Early retention programs are critical.
- **Charges:** High-charge customers churn more. Pricing adjustments or bundled offers may reduce attrition.
- **Contracts:** Long-term contracts are the strongest churn deterrent. Incentives for one-year and two-year contracts should be prioritized.
- **Payment Method:** No significant impact on churn, resources should be focused elsewhere.

8. Recommendations

1. **Retention Programs for New Customers:**
   - Offer onboarding support, loyalty rewards, or discounts within the first 6 months.
2. **Pricing Strategy Adjustments:**
   - Introduce tiered pricing or bundled services to reduce churn among high-charge customers.
3. **Promote Long-Term Contracts:**
   - Provide financial incentives (discounts, loyalty points) for one-year and two-year contracts.
4. **Monitor Predictive Model:**

- Update churn prediction quarterly to refine strategies and adapt to market changes.

9. Conclusion

This analysis demonstrates that churn is primarily driven by tenure, monthly charges, and contract type. By focusing on early retention, pricing strategies, and contract incentives, the business can significantly reduce churn and improve customer lifetime value.